Program 1

```
public class A_SingleDimensionalArray {
        public static void main(String args[]) {

                int intArray2[];
                intArray2 = new int[4];

                int intArray[] = new int[5];

                intArray[0] = 10;
                intArray[1] = 20;
                intArray[2] = 30;
                intArray[3] = 40;
                intArray[4] = 50;


                //int[] original = {10, 9, 8, 7};
                //int[] original = new int[]{10, 9, 8, 7};
                int intArray3[] = {10, 9, 8, 7};

                int array[] = {33, 3, 4, 5};

                System.out.println("\nint Array..."+ intArray3);

                //0,1,2,3 i = 4  -- 4<4 false
                for (int i = 0; i < intArray3.length; i++) {
                   System.out.print(intArray3[i] + " ");
                }

                System.out.println("\n printing Array name using reference
variable "+intArray3 );
                System.out.println("\n using toString" +
Arrays.toString(intArray3));

                System.out.println("\n printing array size  " +
intArray3.length);

                System.out.println("\n printing array size   " + intArray.length);

                System.out.println("Printing array using for each loop");
                for (int i : array)
                {
                   System.out.println(i);
```

```
            }
        }

}
```

Program 2

```
class Man {

}

class B_ArrayDefaultValues
{
   public static void main(String args[])
   {
      int intArray[] = new int[5];

      int intArray2[];
      intArray2 = new int[4];

      int intArray3[] = {10, 9, 8, 7};

      Man manArray[] = new Man[5];

      char charArray[] = new char[5];
      double doubleArray[] = new double[5];
      boolean booleanArray[] = new boolean[5];
      String strArray[] = new String[5];

      System.out.println("int Array...");
      for (int i = 0; i < 5; i++) {
         System.out.print(intArray[i] + " ");
      }

      System.out.println("\n");
      System.out.println("char Array...");
      for (int i = 0; i < 5; i++) {
         System.out.print((int) charArray[i] + " ");
      }

      System.out.println("\n");
```

```
        System.out.println("double Array...");
        for (int i = 0; i < 5; i++) {
            System.out.print(doubleArray[i] + " ");
        }

        System.out.println("\n");
        System.out.println("boolean Array...");
        for (int i = 0; i < 5; i++) {
            System.out.print(booleanArray[i] + " ");
        }

        System.out.println("\n");
        System.out.println("String Array...Predefined class");
        for (int i = 0; i < 5; i++) {
            System.out.print(strArray[i] + " ");
        }

        System.out.println(" \n man array...User defined class");
        for (int i = 0; i < 5; i++) {
            System.out.print(manArray[i] + " ");
        }

    }
}

// Array itself object, so java initializes default values in heap memory
// default value for character is - \u0000
```

Program 3

```
import java.util.Scanner;
import java.util.Arrays;


public class C_TwoDimArray {


    public static void main(String args[]) {


        Scanner scanner = new Scanner(System.in);
```

```
int[][] tArray = new int[2][2]; // 2 rows and 2 columns

int arr[][] = {{1, 2, 3}, {2, 4, 5}, {4, 4, 5}};



//printing 2D array
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      System.out.print(arr[i][j] + " ");
   }
   System.out.println();
}

// Taking input from the user for tArray
System.out.println("Enter 4 elements for the 2x2 array");
for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      System.out.print("Enter element at position [" + i + "][" + j + "]:
");
      tArray[i][j] = scanner.nextInt();
   }
}

System.out.println("\nThe 2x2 array is:");
for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      System.out.print(tArray[i][j] + " ");
   }
   System.out.println(); // New line after each row
```

```
    }

    System.out.println("two dimensional array length");

    System.out.println("Row count" + tArray.length);
    System.out.println("columncount" + tArray[0].length);

    for (int i = 0; i < tArray.length; i++) //row count
    {
       for (int j = 0; j < tArray[i].length; j++) //column count in first row
       {
          System.out.print(tArray[i][j]);
          System.out.print("  ");
       }

       System.out.println();
    }

    System.out.println("printing 2d array using variable " +
Arrays.deepToString(tArray));

    System.out.println("Printing 2 d array using for each");
    for (int[] row : tArray) {
       for (int num : row) {
          System.out.print(num + " ");
       }
       System.out.println(); // New line after each row
    }
```

```
    }


}
```

Program 4

```
public class D_AssigningArray {

    public static void main(String[] args) {
        int[] i = new int[3]; //111
        int[] j;// i want to take the data from i array

        i[0] = 1;
        i[1] = 2;
        i[2] = 3;


        j = i;//assigns the refernce of i to j
        //j = 111

        System.out.println("\n");
        for (int no = 0; no < 3; no++) {
            System.out.println(j[no] + " ");
        }

        //before chnaging the value of j array 0th index = 1
        j[0] =  100;

        System.out.println("\n");
        for (int no = 0; no < 3; no++) {
            System.out.println(j[no] + " ");
        }



        System.out.println("\n");
        for (int no = 0; no < 3; no++) {
            System.out.println(i[no] + " ");
        }

    }
```

}

Decision making stmts

```
//if
public class A_ATMMain
{

     public static void main(String[] args)
     {
              int balance = 10000;
             int withdrawAmount = 5000;

             if(withdrawAmount <= balance)
             {
                 System.out.println("Withdrawal
Successful");
             }
     }

}
```

```
public class B_UserMain {

      public static void main(String[] args) {

             String enteredPassword = "java123";

             if(enteredPassword.equals("java123"))
             {
                 System.out.println("Login Successful");
             }
             else
             {
                 System.out.println("Invalid Password");
             }
         }
```

```
}
```

```
//if else if ladder
public class C_IncomeTaxMain {

	public static void main(String[] args) {

		double salary = 900000;

		if(salary >= 1500000)
		{
			System.out.println("Tax Slab : 30%");
		}
		else if(salary >= 700000)
		{
			System.out.println("Tax Slab : 20%");
		}
		else if(salary >= 300000)
		{
			System.out.println("Tax Slab : 10%");
		}
		else
		{
			System.out.println("No Tax");
		}

	}

}
```

```
//nested if
public class D_BankingMain {

	public static void main(String[] args) {

		boolean accountActive = true;
		double balance = 15000;
		double transferAmount = 5000;

		if(accountActive)
		{
```

```
            if(transferAmount <= balance)
            {
                System.out.println("Money Transfer
Successful");
            }
            else
            {
                System.out.println("Insufficient
Balance");
            }
        }
        else
        {
            System.out.println("Account is Inactive");
        }
    }
}
```

```
import java.util.Scanner;
//switch case choice is int
public class E_FoodMain {
            public static void main(String[] args) {

                    Scanner scanner = new Scanner(System.in);

                    System.out.println("----- Food Menu -----
");
                    System.out.println("1. Idli");
                    System.out.println("2. Dosa");
                    System.out.println("3. Fried Rice");
                    System.out.println("4. Meals");

                    System.out.print("Enter your choice: ");
                    int choice = scanner.nextInt();//1

                    switch(choice)//1
                    {
                        case 1:
                            System.out.println("Idli
Ordered");
                            break;
```

```
                    case 2:
                        System.out.println("Dosa
Ordered");
                        break;

                    case 3:
                        System.out.println("Fried Rice
Ordered");
                        break;

                    case 4:
                        System.out.println("Meals
Ordered");
                        break;

                    default:
                        System.out.println("Invalid
Choice");
                }

                scanner.close();

    }

}
```

```
import java.util.Scanner;

public class F_GradeMain {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter Grade (A/B/C): ");
        //scanner.next() always returns a String, not a
char.
        char grade = scanner.next().charAt(0);

        switch (grade) {
```

```
            case 'A':
                System.out.println("Excellent");
                break;

            case 'B':
                System.out.println("Good");
                break;

            case 'C':
                System.out.println("Average");
                break;

            default:
                System.out.println("Invalid Grade");
        }

        scanner.close();
    }
}
```

Iteration

```
public class A_ForLoop {

    public static void main(String[] args) {

                for(int i = 1; i <= 5; i++)
                {
                    System.out.println(i);
                }
            }
}
```

```
public class B_WhileLoop {

    public static void main(String[] args) {
```

```
            int i = 1;

            while(i <= 5)//6<=5
            {
                System.out.println(i);
                i++;
            }

    }

}
```

```
public class C_DoWhileLoop {

    public static void main(String[] args) {

        int i = 1;

        do
        {
            System.out.println(i);
            i++;
        }
        while(i <= 5);
    }
}
```

```
public class D_WhileDoWhile {
      public static void main(String[] args) {
          System.out.println("While loop");
            int i = 6;

            while(i <= 5)
            {
                System.out.println(i);
                i++;
            }

            System.out.println("do While loop");
            int k = 6;

            do
```

```
            {
                System.out.println(k);
                k++;
            }
            while(k <= 5);
        }
}
```

```
import java.util.Scanner;

public class E_SumApp {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        char choice;

        do {

            System.out.print("Enter First Number: ");
            int num1 = scanner.nextInt();

            System.out.print("Enter Second Number: ");
            int num2 = scanner.nextInt();

            int sum = num1 + num2;

            System.out.println("Sum = " + sum);

            System.out.print("Do you want to perform the
operation again? (Y/N): ");
            choice = scanner.next().charAt(0);

        } while (choice == 'Y' || choice == 'y');

        System.out.println("Thank You!");

        scanner.close();
    }
}
```